SLIC Index V3

Methodology Technical Paper

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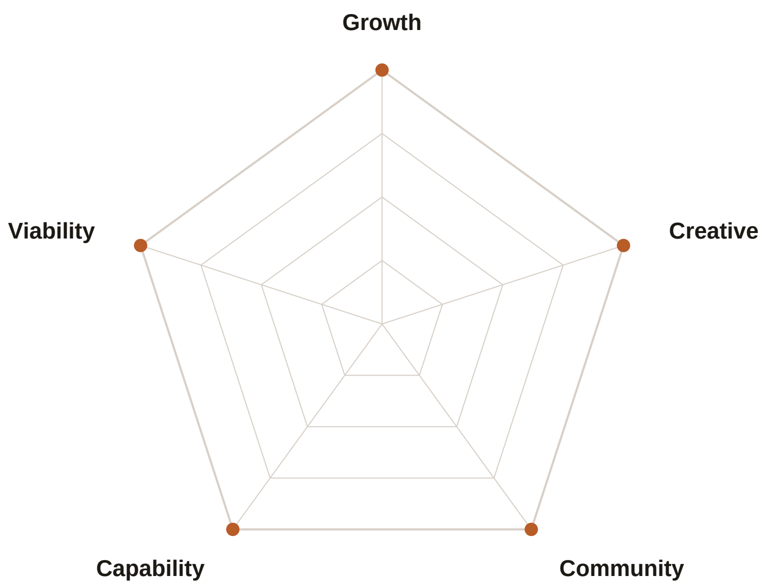
This paper documents the complete scoring methodology for the SLIC Index V3: indicator definitions, normalization anchor tables, aggregation formula, coverage grades, and the full published rankings for 157 cities.



Five pillars. Fifty-point symmetry. Absolute anchors.

Executive Summary

The Smart and Liveable Cities Index (SLIC) V3 is a transparent, open-source city ranking system measuring quality of life for people in the process of building lives — not tourists, not expats on hardship pay, not global capital seeking returns. It covers 157 published cities across an Asia-Pacific-centred but globally inclusive dataset, scored on 35 indicators grouped into five pillars.

Five pillars:

Pillar	Weight	What it measures
Growth	25%	Economic dynamism, disposable income, startup density, civic freedom
Viability	22%	Safety, air quality, healthcare, housing affordability, water access
Capability	18%	Transit, digital infrastructure, education, renewable energy, walkability
Community	15%	Tolerance, LGBTQ+ rights, gender equality, social trust, income inequality
Creative	20%	Cultural venues, culinary diversity, arts funding, creative employment

Absolute scoring philosophy:

- All indicators are normalized against fixed absolute benchmarks — adding city #501 never changes any existing score.
- Higher normalized scores always mean better urban outcomes; harmful indicators are reverse-scored inside the normalization step.
- The Adjusted Mazziotta–Pareto Index (AMPI) aggregation formula penalizes pillar imbalance: a city with one excellent pillar and four poor ones scores lower than a city with five moderate pillars.
- Missing data is never imputed. Cities with fewer than the minimum required indicators receive a coverage penalty (–5 or –15 points) and a visible grade flag.
- Every source is named. Every score is traceable back to a raw data point, a normalization function, and a published source.

1.

Scoring Framework

1.1 Piecewise Linear Normalization

Each raw metric value is mapped to a 0–100 score using a fixed set of absolute anchor points. Between anchor points, scores are linearly interpolated. Values beyond the outer anchors are clamped to 0 or 100. For lower-is-better indicators the anchor mapping is reversed so that lower raw values produce higher normalized scores.

```
s_m(c) = piecewise_linear(x_m(c), [(t_0, 0), (t_1, 25), (t_2, 50), (t_3, 75), (t_4, 100)])

Where:
  x_m(c)   = raw metric value for city c on metric m
  t_0..t_4 = fixed absolute anchor thresholds (see indicator chapters)
  s_m(c)   = normalized metric score, clamped to [0, 100]
```

1.2 Pillar Aggregation (AMPI)

Within each pillar, constituent metric scores are aggregated using the Adjusted Mazziotta–Pareto Index (AMPI). The AMPI penalizes uneven performance: a city with high variance across its metrics scores lower than one with similar mean but lower variance. This makes it impossible to compensate for a catastrophic metric score by excelling elsewhere.

```
μ    = (1/n) × Σ s_m(c)                  [arithmetic mean]
σ    = √((1/n) × Σ (s_m(c) − μ)²)        [population standard deviation]
cv   = σ / μ                               [coefficient of variation]

AMPI = μ − (σ × cv)  =  μ − σ²/μ

pillar_score = max(0, AMPI − coverage_penalty)
```

1.3 Overall SLIC Score

```
SLIC(c) = AMPI applied across five pillar scores

μ_p  = (G×0.25 + V×0.22 + Cap×0.18 + Com×0.15 + Cr×0.20) / 1.00

Then AMPI(μ_p, σ_p) is applied once more to penalise
cities with extreme pillar imbalance.
```

1.4 PPP-Adjusted Disposable Income (DI_PPP)

The signature SLIC metric. Measures residual monthly income after all essential costs, converted to PPP-adjusted USD so cities in different economic contexts are comparable.

```
DI_PPP(c) = [
    GrossIncome(c) × (1 − TaxRate(country(c)))
    − Rent(c)
    − Utilities(c)
    − Transit(c)
    − Internet(c)
    − Food(c)
  ] ÷ PPP_PrivateConsumptionFactor(country(c))
```

2.

Pillar 1: Growth

Economic dynamism and opportunity (25% weight)

The Growth pillar measures whether a city creates the conditions for people to build economically productive lives: disposable income after costs, startup and innovation density, civic freedom, and broad economic momentum.

G1 — Real GDP Growth (5-year avg)

Unit: % per annum · Direction: ↑ Higher is better · Source: IMF World Economic Outlook; OECD Regional Statistics

Raw value	Score
≤ 0	0
1.0	25
2.5	50
4.0	75
≥ 6.0	100

G2 — Startup Density

Unit: per 100,000 population · Direction: ↑ Higher is better · Source: Crunchbase; Dealroom

Raw value	Score
≤ 5	0
20	25
50	50
100	75
≥ 200	100

G3 — VC Investment Intensity

Unit: USD per capita, 3-year avg · Direction: ↑ Higher is better · Source: Crunchbase; PitchBook

Raw value	Score
≤ 5	0
50	25
200	50
500	75
≥ 1,500	100

G4 — Ease of Doing Business

Unit: 0–100 index · Direction: ↑ Higher is better · Source: World Bank B-READY Index

Direct passthrough; no transformation required.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

G5 — Civic Freedom

Unit: 0–100 composite · Direction: ↑ Higher is better · Source: Freedom House (0.6) + V-Dem Liberal Democracy Index (0.4)

Composite: 0.6 × FH_rescaled(0–100) + 0.4 × V-Dem_rescaled(0–100). Direct passthrough after composite.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

G6 — Patent Applications

Unit: per 100,000 population, 3-year avg · Direction: ↑ Higher is better · Source: World Intellectual Property Organization (WIPO)

Raw value	Score
≤ 2	0
15	25
40	50
80	75
≥ 150	100

G7 — High-Skill Employment

Unit: % ISCO categories 1–3 · Direction: ↑ Higher is better · Source: International Labour Organization (ILO); LinkedIn Workforce Insights

Raw value	Score
≤ 10	0
20	25
30	50
40	75
≥ 55	100

3.

Pillar 2: Viability

Lived sustainability and safety (22% weight)

The Viability pillar measures whether a city is liveable at the ground level: personal safety, air quality, healthcare, housing cost burden, water access, climate stability, and demographic vitality.

V1 — Homicide Rate

Unit: per 100,000 population · Direction: ↓ Lower is better · Source: UNODC International Homicide Statistics

Lower is better; anchor order is reversed.

Raw value	Score
≥ 50	0
20	25
5	50
1.0	75
≤ 0.3	100

V2 — Air Quality (PM2.5)

Unit: µg/m³ annual mean · Direction: ↓ Lower is better · Source: WHO Global Air Quality Database; IQAir World Air Quality Report

Lower is better; reversed anchors.

Raw value	Score
≥ 80	0
50	25
25	50
10	75
≤ 5	100

V3 — Healthcare Access & Quality

Unit: HAQ Index 0–100 · Direction: ↑ Higher is better · Source: IHME / Lancet Healthcare Access and Quality Index

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

V4 — PPP-Adjusted Disposable Income

Unit: USD/month residual · Direction: ↑ Higher is better · Source: Derived: Numbeo cost data + World Bank PPP factors

Non-linear anchors reflect diminishing returns above USD 2,000/month.

Raw value	Score
≤ 0	0
200	15
500	35
1,000	55
2,000	75
≥ 4,000	100

V5 — Housing Burden

Unit: % of gross income · Direction: ↓ Lower is better · Source: Numbeo housing cost surveys; OECD Affordable Housing Database

Lower burden = higher score.

Raw value	Score
≥ 70	0
50	25
30	50
20	75
≤ 10	100

V6 — Water & Sanitation

Unit: % population with safely managed access · Direction: ↑ Higher is better · Source: WHO/UNICEF Joint Monitoring Programme

Raw value	Score
≤ 50	0
70	25
85	50
95	75
≥ 99	100

V7 — Peace & Stability

Unit: Global Peace Index (inverted) · Direction: ↑ Higher is better · Source: Institute for Economics and Peace (IEP)

Formula: score = max(0, min(100, (4 – GPI) / 3 × 100)).

Raw value	Score
GPI ≥ 3.5 → 0	0
GPI 3.0 → 17	~17
GPI 2.0 → 50	50
GPI 1.5 → 83	~83
GPI ≤ 1.2 → 100	100

V8 — Birth Rate

Unit: per 1,000 population · Direction: ↑ Higher is better · Source: UN Population Division; national statistics offices

Interpreted as demographic optimism signal, not population policy judgment.

Raw value	Score
≤ 4	0
7	25
10	50
14	75
≥ 20	100

4.

Pillar 3: Capability

Infrastructure and systems quality (18% weight)

The Capability pillar measures urban infrastructure quality: transit coverage, digital connectivity, digital governance, education quality, walkability, cycling infrastructure, and renewable energy share.

C1 — Transit Coverage

Unit: % population within 500m of stop · Direction: ↑ Higher is better · Source: GTFS feeds; ITDP Transit-Oriented Development Database

Raw value	Score
≤ 5	0
20	25
40	50
65	75
≥ 85	100

C2 — Internet Speed

Unit: Mbps median fixed broadband · Direction: ↑ Higher is better · Source: Ookla Speedtest Global Index; M-Lab NDT

Raw value	Score
≤ 5	0
25	25
75	50
150	75
≥ 300	100

C3 — Digital Government

Unit: UN E-Government Development Index × 100 · Direction: ↑ Higher is better · Source: UN Department of Economic and Social Affairs

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

C4 — Education Quality

Unit: PISA average score · Direction: ↑ Higher is better · Source: OECD PISA; UNESCO Institute for Statistics (tertiary fallback)

Tertiary enrollment fallback anchors: ≤5%=0, 15%=25, 25%=50, 40%=75, ≥55%=100.

Raw value	Score
≤ 350	0
400	25
450	50
500	75
≥ 550	100

C5 — Walkability

Unit: Walk Score 0–100 · Direction: ↑ Higher is better · Source: Walk Score; OpenStreetMap pedestrian network analysis

Direct passthrough.

Raw value	Score
0	0
25	25
50	50
75	75
100	100

C6 — Cycling Infrastructure

Unit: km of protected lanes per 100,000 pop · Direction: ↑ Higher is better · Source: OpenStreetMap CyclOSM layer; Copenhagenize Index

Raw value	Score
≤ 1	0
5	25
15	50
30	75
≥ 50	100

C7 — Renewable Energy Share

Unit: % of electricity from renewables · Direction: ↑ Higher is better · Source: CDP Cities; IRENA; national grid operators

Raw value	Score
≤ 2	0
15	25
35	50
60	75
≥ 90	100

5.

Pillar 4: Community

Belonging, tolerance, and social equity (15% weight)

The Community pillar measures whether a city is genuinely open: LGBTQ+ rights, religious freedom, immigrant integration, income inequality, social trust, gender equality, and weekend retention (a proxy for residents choosing to stay).

O1 — LGBTQ+ Rights

Unit: Composite 0–100 · Direction: ↑ Higher is better · Source: ILGA World; Equaldex

Composite: same-sex marriage (25pts) + anti-discrimination law (25pts) + gender recognition (15pts) + no criminalization (20pts) + social acceptance survey (15pts).

Raw value	Score
0	0
25	25
50	50
75	75
100	100

O2 — Religious Freedom

Unit: Pew GRI inverted · Direction: ↑ Higher is better · Source: Pew Research Center Global Restrictions Index

Formula: score = max(0, min(100, (10 – GRI) / 10 × 100)).

Raw value	Score
GRI ≥ 9 → 0	0
GRI 7 → 22	~22
GRI 5 → 44	~44
GRI 2 → 78	~78
GRI ≤ 0.5 → 95	~95

O3 — Immigrant Integration

Unit: Employment gap (percentage points) · Direction: ↓ Lower is better · Source: OECD International Migration Outlook; national labour force surveys

Lower employment gap = higher score.

Raw value	Score
≥ 30 pp	0
20 pp	25
10 pp	50
5 pp	75
≤ 1 pp	100

O4 — Income Inequality (Gini)

Unit: Gini coefficient 0–1 · Direction: ↓ Lower is better · Source: World Bank PovcalNet; OECD Income Distribution Database

Lower Gini = higher score.

Raw value	Score
≥ 0.60	0
0.45	25
0.35	50
0.28	75
≤ 0.22	100

O5 — Social Trust

Unit: % saying most people can be trusted · Direction: ↑ Higher is better · Source: World Values Survey; Gallup World Poll

Raw value	Score
≤ 5%	0
15%	25
30%	50
50%	75
≥ 70%	100

O6 — Gender Equality

Unit: UNDP GII inverted · Direction: ↑ Higher is better · Source: UNDP Human Development Reports — Gender Inequality Index

Formula: score = (1 – GII) × 100.

Raw value	Score
GII 1.0 → 0	0
GII 0.75 → 25	25
GII 0.5 → 50	50
GII 0.25 → 75	75
GII 0.0 → 100	100

O7 — Weekend Retention

Unit: Weekend / weekday mobility ratio · Direction: ↑ Higher is better · Source: Mobile analytics platforms; city-level aggregated movement data

Ratio above 1.0 indicates residents actively choose to stay or return on weekends.

Raw value	Score
≤ 0.70	0
0.80	25
0.90	50
1.00	75
≥ 1.10	100

6.

Pillar 5: Creative

Cultural richness and creative economy (20% weight)

The Creative pillar measures lived cultural richness for residents, not tourists: cultural venue density, UNESCO heritage access, culinary diversity, nightlife, arts funding, creative employment share, and international event density.

R1 — Cultural Venues

Unit: per 100,000 population · Direction: ↑ Higher is better · Source: OpenStreetMap; Google Places API

Raw value	Score
≤ 2	0
8	25
20	50
40	75
≥ 70	100

R2 — UNESCO World Heritage Sites

Unit: sites within 50km radius · Direction: ↑ Higher is better · Source: UNESCO World Heritage List + GIS distance analysis

Raw value	Score
0	0
1	25
3	50
6	75
≥ 10	100

R3 — Culinary Diversity

Unit: cuisine types per 100,000 population · Direction: ↑ Higher is better · Source: Google Places; Yelp; Foursquare

Raw value	Score
≤ 1	0
3	25
8	50
15	75
≥ 25	100

R4 — Nightlife Density

Unit: bars and clubs per 100,000 population · Direction: ↑ Higher is better · Source: OpenStreetMap; Google Places

Raw value	Score
≤ 3	0
10	25
25	50
50	75
≥ 80	100

R5 — Arts Funding

Unit: USD PPP per capita public arts expenditure · Direction: ↑ Higher is better · Source: Eurostat Culture Statistics; national ministries of culture

Raw value	Score
≤ 5	0
30	25
80	50
150	75
≥ 300	100

R6 — Creative Employment

Unit: % of workforce in creative industries · Direction: ↑ Higher is better · Source: UNCTAD Creative Economy Report; national labour force surveys

Raw value	Score
≤ 1%	0
3%	25
6%	50
10%	75
≥ 15%	100

R7 — International Events

Unit: per million population per year · Direction: ↑ Higher is better · Source: ICCA International Congress and Convention Association; Eventbrite

Raw value	Score
≤ 2	0
10	25
25	50
50	75
≥ 100	100

7.

Data Sources

SLIC uses a five-tier source hierarchy. Tier 1 (city-official) is preferred; the fallback order is Tier 2 (subnational) → Tier 3 (national/international) → Tier 4 (audited secondary) → Tier 5 (analyst assessment). The source tier for each published metric is visible on the city scorecard page.

Tier	Level	Examples
1	City / metro official	Municipal open data portals, utility feeds, transit GTFS
2	Subnational official	State, provincial, or regional government data
3	National / international official	World Bank, WHO, ILO, UNESCO, OECD, WIPO, UN Agencies
4	Audited secondary & experimental	OpenAQ, Copernicus CAMS, M-Lab NDT, satellite remote sensing
5	Analyst assessment	SLIC analyst cross-reference and lived-experience review

Primary Source Organisations

- World Bank — WDI, B-READY, PovcalNet, PPP conversion factors
- World Health Organization — GHO OData API, HAQ Index, JMP water/sanitation
- International Labour Organization — ILOSTAT (employment, hours, wages)
- OECD — PISA, Health Statistics, Affordable Housing, Income Distribution
- UN Population Division — birth rates, demographic projections
- UNESCO Institute for Statistics — education enrollment, cultural indicators
- UNODC — International Homicide Statistics
- UNDP — Human Development Reports, Gender Inequality Index
- IEP — Global Peace Index
- Freedom House — Freedom in the World annual scores
- V-Dem Institute — Liberal Democracy Index
- Pew Research Center — Global Restrictions on Religion Index
- WIPO — Patent Application Statistics
- IMF — World Economic Outlook GDP data
- IRENA — Renewable capacity and generation statistics
- IQAir / WHO — PM2.5 ambient air quality annual averages
- ILGA World + Equaldex — LGBTQ+ legal environment composite
- World Values Survey + Gallup — social trust indicators
- ICCA — International Congress and Convention Association statistics
- Numbeo — Cost of living, housing, and consumer price surveys
- Crunchbase / PitchBook / Dealroom — startup and VC investment data
- Ookla + M-Lab — broadband speed measurements
- OpenStreetMap / CyclOSM — cycling and pedestrian infrastructure
- Walk Score — walkability index

- OpenAQ / Copernicus CAMS — supplementary air quality monitoring

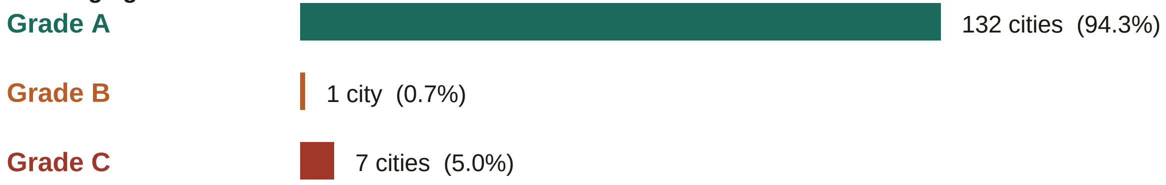
8.

Coverage Grades

SLIC does not impute missing values. When indicators are unavailable, they are excluded from the pillar aggregation and a coverage penalty is applied. The coverage grade is published alongside every city score.

Grade	Condition	Penalty	Interpretation
A	≥ 6 of 7 indicators present (or ≥ 5 of 5–6 indicator pillar)	None	Full confidence in pillar score
B	≥ 4 indicators present	–5 points	Score is reliable but interpret with moderate caution
C	≤ 3 indicators present	–15 points, flagged provisional	Treat as indicative; significant data gaps exist

Coverage grade distribution across 157 cities



The penalty is applied after the AMPI computation at the pillar level and is floored at zero. Overall coverage grade reflects the worst pillar grade for that city. Cities are never hidden because of low coverage — low coverage is surfaced, not suppressed.

9.

Published City Rankings

All 157 published cities, sorted by overall SLIC score. Scores are rounded to one decimal place. Ties are allowed — cities with the same rounded score share the same rank number. Pillar column headers: G = Growth, V = Viability, Cap = Capability, Com = Community, Cr = Creative.

#	City	Country	SLIC	G	V	Cap	Com	Cr	Gr.
1	Kaohsiung	Taiwan	67.6	52.4	87.4	90.6	70.2	42.0	A
2	Taipei	Taiwan	66.9	40.5	85.1	89.7	68.1	58.7	A
3	Raleigh	United States	64.0	54.8	80.7	79.1	39.3	62.0	A
4	Busan	South Korea	63.7	47.2	92.3	85.8	40.5	50.6	A
4	Katowice	Poland	63.7	65.7	79.5	65.1	64.9	41.9	A
6	Fukuoka	Japan	62.4	50.7	92.0	67.4	46.0	52.3	A
7	Bangkok	Thailand	61.8	56.4	85.7	61.4	88.3	22.8	A
8	Lyon	France	61.6	42.2	84.3	63.9	73.7	50.1	A
8	Montreal	Canada	61.6	51.6	70.4	75.3	58.7	54.1	A
10	Santiago	Chile	61.3	54.0	78.6	83.8	43.0	44.6	A
11	Pittsburgh	United States	61.2	56.3	66.3	79.1	39.3	62.0	A
12	Kobe	Japan	61.1	49.5	94.6	67.4	42.0	47.3	A
13	Minneapolis	United States	60.6	52.4	68.1	79.1	39.3	62.0	A
14	Haifa	Israel	60.3	54.5	89.2	64.7	29.8	54.7	A
15	Suwon	South Korea	60.2	41.3	86.1	85.8	35.8	50.6	A
16	Ottawa	Canada	60.0	47.0	68.6	75.3	58.7	54.1	A
17	Valparaiso	Chile	59.8	55.0	72.7	83.8	40.0	44.6	A
18	Milan	Italy	59.5	72.2	76.4	66.9	45.0	29.4	A
19	Eindhoven	Netherlands	59.3	53.2	75.1	71.8	53.2	42.8	A
20	Graz	Austria	59.2	47.3	71.8	74.4	66.2	41.4	A
21	Braga	Portugal	59.1	52.5	84.6	68.8	41.7	43.8	A
21	Tel Aviv	Israel	59.1	46.2	91.0	64.7	32.8	54.7	A
23	Chicago	United States	58.9	42.5	71.7	79.1	39.3	62.0	A
24	Porto	Portugal	58.8	48.2	88.2	68.8	41.7	43.8	A
25	Tallinn	Estonia	58.7	47.5	59.3	58.4	58.1	72.8	A
25	Torun	Poland	58.7	62.3	79.5	65.1	49.9	31.9	A
27	Incheon	South Korea	58.6	41.3	86.0	85.8	25.5	50.6	A
27	Singapore	Singapore	58.6	35.2	74.8	94.1	11.3	73.3	A
29	Zurich	Switzerland	58.4	32.3	78.6	72.7	66.8	49.4	A
30	Copenhagen	Denmark	58.3	37.6	61.8	71.3	67.8	61.5	A
31	Khobar	Saudi Arabia	58.0	75.0	96.3	78.9	17.4	6.1	A
31	Sydney	Australia	58.0	22.4	94.1	87.1	40.5	49.7	A
33	Jeddah	Saudi Arabia	57.6	73.7	96.3	78.9	17.4	6.1	A
34	Brisbane	Australia	57.5	28.8	94.1	87.1	37.5	41.7	A
35	Vienna	Austria	57.2	37.5	73.6	74.4	66.2	41.4	A
36	Melbourne	Australia	56.9	30.8	82.9	87.1	39.5	46.7	A
36	Perth	Australia	56.9	28.9	94.1	87.1	35.5	39.7	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Gr.
38	Manama	Bahrain	56.8	70.1	52.6	57.8	41.7	55.5	A
39	Adelaide	Australia	56.3	31.1	91.9	87.1	35.5	36.7	A
40	Toronto	Canada	56.1	28.2	72.2	75.3	58.7	54.1	A
41	San Juan	Puerto Rico	55.9	48.9	59.2	68.3	61.2	45.8	B
42	Amsterdam	Netherlands	55.7	38.8	75.1	71.8	53.2	42.8	A
43	Tianjin	China	55.5	57.0	76.7	71.9	26.5	37.1	A
44	Hiroshima	Japan	55.4	45.7	94.6	67.4	24.0	37.3	A
44	Paris	France	55.4	28.9	78.9	63.9	68.7	45.1	A
46	Gdansk	Poland	55.2	57.3	61.5	65.1	54.9	36.9	A
47	Vancouver	Canada	55.0	22.3	74.0	75.3	58.7	54.1	A
48	Bratislava	Slovakia	54.7	54.7	75.3	49.7	56.2	35.3	A
48	Jeju City	South Korea	54.7	35.1	88.7	85.8	25.5	35.6	A
48	Krakow	Poland	54.7	52.3	65.1	65.1	54.9	36.9	A
51	Chiang Mai	Thailand	54.6	61.9	60.7	56.4	73.9	22.8	A
51	Port Louis	Mauritius	54.6	66.0	76.2	47.0	25.3	45.4	A
53	Bologna	Italy	54.1	49.2	77.9	66.9	45.0	29.4	A
54	Kota Kinabalu	Malaysia	53.8	67.3	75.4	52.4	31.4	31.5	A
55	Kuching	Malaysia	53.7	67.3	73.7	52.4	32.6	31.5	A
56	Taichung	Taiwan	53.2	67.8	—	—	0.0	74.7	Watchlist
57	Melaka	Malaysia	53.1	68.2	70.5	52.4	31.7	31.5	A
58	George Town	Malaysia	53.0	64.7	73.7	52.4	32.6	31.5	A
58	Sapporo	Japan	53.0	45.7	76.7	67.4	34.0	37.3	A
60	Salalah	Oman	52.6	69.9	71.1	42.0	38.5	30.6	A
61	Cordoba	Argentina	52.3	59.5	69.2	78.6	40.6	10.1	A
61	Dubai	United Arab Emirates	52.3	55.8	61.9	62.0	48.9	31.2	A
61	Guangzhou	China	52.3	41.7	88.0	71.9	14.5	37.1	A
64	Abu Dhabi	United Arab Emirates	52.2	56.9	60.2	62.0	48.9	31.2	A
65	Budapest	Hungary	52.0	41.5	79.3	55.5	59.6	26.2	A
65	Montevideo	Uruguay	52.0	50.5	61.5	74.6	50.5	24.3	A
67	Hangzhou	China	51.9	42.4	76.7	71.9	27.4	37.1	A
68	Auckland	New Zealand	51.7	20.2	82.6	85.7	40.7	35.1	A
69	Shenzhen	China	51.6	36.5	92.5	71.9	11.5	37.1	A
70	Christchurch	New Zealand	51.4	25.7	80.8	85.7	38.7	30.1	A
71	Muscat	Oman	51.2	64.3	71.1	42.0	38.5	30.6	A
72	Male	Maldives	50.9	64.4	72.1	39.8	31.2	35.4	A
73	San Jose	Costa Rica	50.8	49.6	56.1	66.0	46.7	35.8	A
74	Buenos Aires	Argentina	50.3	49.4	71.4	78.6	40.6	10.1	A
74	Chongqing	China	50.3	49.3	61.5	71.9	27.1	37.1	A
76	Hobart	Australia	50.1	29.1	71.6	87.1	33.5	31.7	A
77	Chengdu	China	50.0	48.2	61.2	71.9	27.3	37.1	A
77	Tbilisi	Georgia	50.0	59.7	58.9	33.1	—	43.1	Watchlist
79	Dunedin	New Zealand	49.4	27.6	73.6	85.7	36.7	27.1	A
79	Kigali	Rwanda	49.4	76.2	68.0	15.8	50.7	24.8	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Gr.
81	Wellington	New Zealand	49.2	20.0	77.2	85.7	38.7	30.1	A
82	Casablanca	Morocco	48.8	60.0	70.3	34.1	26.1	41.2	A
83	Shanghai	China	48.7	33.0	83.5	71.9	11.5	37.1	A
84	Bucharest	Romania	48.6	50.8	71.7	50.6	26.8	34.8	A
85	Kuala Lumpur	Malaysia	48.5	59.7	68.5	52.4	18.6	31.5	A
86	Helsinki	Finland	48.4	29.8	47.4	81.0	35.3	52.9	A
86	Kampala	Uganda	48.4	54.6	—	68.7	—	22.4	Watchlist
88	Belgrade	Serbia	48.3	51.4	67.5	53.2	33.3	30.3	A
88	Cuenca	Ecuador	48.3	58.6	—	78.8	—	8.0	Watchlist
90	Panama City	Panama	48.0	59.1	52.7	64.2	31.5	26.9	A
91	Curitiba	Brazil	47.3	59.5	49.9	69.5	22.9	27.5	A
91	Riyadh	Saudi Arabia	47.3	72.3	54.2	78.9	15.4	4.1	A
93	Medellin	Colombia	46.8	55.6	53.2	66.3	35.6	19.6	A
94	Asuncion	Paraguay	46.7	52.0	—	74.7	—	14.9	Watchlist
94	Rabat	Morocco	46.7	61.0	59.9	34.1	26.1	41.2	A
96	Valencia	Spain	46.5	43.6	100.0	31.1	12.9	30.4	C
97	Hat Yai	Thailand	46.2	50.2	62.8	56.4	33.9	22.8	A
98	Da Nang	Vietnam	46.1	69.6	—	44.3	43.0	20.7	C
98	Nizhny Novgorod	Russia	46.1	62.4	62.4	59.2	20.1	15.5	A
100	Phuket	Thailand	46.0	44.7	68.4	56.4	33.9	22.8	A
101	Florianopolis	Brazil	45.9	58.1	45.1	69.5	22.9	27.5	A
102	Sao Paulo	Brazil	45.8	50.3	53.5	69.5	22.9	27.5	A
103	Amman	Jordan	45.7	48.6	71.6	46.5	8.5	40.8	A
104	Aqaba	Jordan	45.4	52.2	66.0	46.5	8.5	40.8	A
105	Zagreb	Croatia	44.1	60.5	48.6	17.8	—	42.3	Watchlist
106	Phnom Penh	Cambodia	43.1	66.5	—	18.2	—	36.3	Watchlist
107	Dar es Salaam	Tanzania	42.8	54.6	—	59.8	—	12.7	Watchlist
108	Arequipa	Peru	42.6	67.7	34.1	42.3	37.6	24.7	A
109	Surabaya	Indonesia	42.5	79.4	40.8	35.1	25.6	17.7	A
110	Bogota	Colombia	41.9	50.2	37.0	66.3	35.6	19.6	A
111	Cebu City	Philippines	41.8	74.2	38.2	40.4	27.0	17.5	A
112	Nairobi	Kenya	41.6	64.3	69.1	12.5	34.2	14.5	A
113	Pune	India	41.4	58.6	67.7	30.4	23.2	14.6	A
114	Lima	Peru	41.0	61.3	34.1	42.3	37.6	24.7	A
115	Makati	Philippines	40.8	70.2	38.2	40.4	27.0	17.5	A
116	Kuwait City	Kuwait	40.7	65.0	45.0	65.0	16.7	1.9	A
117	Moscow	Russia	40.6	40.6	62.4	59.2	20.1	15.5	A
118	Hyderabad	India	40.3	58.7	62.3	30.4	23.2	14.6	A
119	Guadalajara	Mexico	40.2	46.1	40.0	55.5	47.0	14.1	A
120	Jakarta	Indonesia	40.0	74.2	45.3	35.1	10.6	17.7	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Gr.
120	Thimphu	Bhutan	40.0	60.6	47.0	25.2	20.1	34.7	A
122	Brno	Czechia	39.8	67.6	38.3	0.0	—	42.3	Watchlist
123	Accra	Ghana	38.9	74.3	44.0	14.9	29.6	17.7	A
124	Vilnius	Lithuania	38.7	62.1	7.5	41.6	—	41.4	Watchlist
125	Kumasi	Ghana	38.3	79.1	35.9	14.9	29.6	17.7	A
126	Alexandria	Egypt	38.2	63.7	—	11.3	—	30.6	Watchlist
127	Bengaluru	India	37.9	57.4	53.1	30.4	23.2	14.6	A
128	Ljubljana	Slovenia	37.6	63.1	43.4	0.0	—	33.2	Watchlist
129	Mexico City	Mexico	37.4	36.4	38.2	55.5	47.0	14.1	A
129	Windhoek	Namibia	37.4	55.3	30.5	32.1	40.1	25.4	A
131	Chandigarh	India	37.2	59.0	47.7	30.4	23.9	14.6	A
132	Doha	Qatar	37.0	44.4	48.0	54.4	11.1	19.6	A
133	Santo Domingo	Dominican Republic	36.9	31.7	—	53.7	—	28.2	Watchlist
134	Kandy	Sri Lanka	35.9	62.7	40.7	33.7	19.5	11.7	A
135	Cape Town	South Africa	35.6	31.6	44.8	26.3	47.3	29.8	A
135	Riga	Latvia	35.6	62.5	2.3	33.5	—	40.3	Watchlist
137	Chattogram	Bangladesh	35.5	65.2	48.3	10.8	23.3	15.7	A
137	Gaborone	Botswana	35.5	49.7	31.9	22.2	28.6	39.0	A
139	Colombo	Sri Lanka	35.4	60.7	40.7	33.7	19.5	11.7	A
140	Merida	Mexico	35.2	49.8	13.3	55.5	47.0	14.1	A
141	Dhaka	Bangladesh	35.1	63.4	48.3	10.8	23.3	15.7	A
142	Johannesburg	South Africa	34.8	31.7	41.2	26.3	47.3	29.8	A
143	Suva	Fiji	34.6	49.6	36.1	47.7	14.8	17.1	A
144	Pokhara	Nepal	31.6	62.5	37.6	12.2	14.8	16.6	A
145	Gothenburg	Sweden	31.5	42.6	7.5	14.5	26.1	63.3	C
146	Kathmandu	Nepal	31.1	60.4	37.6	12.2	14.8	16.6	A
147	Cork	Ireland	30.7	51.4	17.7	13.3	26.9	37.4	C
148	Antwerp	Belgium	29.8	42.6	28.0	4.4	27.7	40.4	C
149	Dakar	Senegal	28.6	58.2	0.7	10.6	44.6	26.7	A
150	Karachi	Pakistan	27.8	59.3	32.7	8.1	22.8	4.3	A
150	Lahore	Pakistan	27.8	59.5	32.7	8.1	22.8	4.3	A
152	Islamabad	Pakistan	27.6	58.6	32.7	8.1	22.8	4.3	A
153	Munich	Germany	26.9	4.5	48.6	27.1	19.7	36.2	C
154	Bergen	Norway	24.7	44.6	0.0	3.2	21.7	48.8	C
155	Port Vila	Vanuatu	20.8	22.1	—	26.7	—	13.9	Watchlist
156	Apia	Samoa	16.5	27.9	—	—	—	2.2	Watchlist
157	Port Moresby	Papua New Guinea	5.7	10.3	—	—	—	0.0	Watchlist

10.

Design Principles

Absolute scoring

Anchors are fixed at real-world thresholds. A city's score does not change because a new city joined the index. Adding city #501 cannot change cities #1–500.

Transparent and traceable

Every published score traces back to a raw data point, a normalization function, and a named source. The scoring workbook is publicly downloadable.

Penalises imbalance

The AMPI aggregation formula means a city cannot compensate for catastrophic performance in one pillar by excelling in another. Extreme imbalance is structurally penalised.

Honest about data gaps

Missing data is never imputed. Gaps are flagged with visible coverage grades and penalties. A low-coverage city is shown, not hidden.

Scalable without revision

New cities can be added to the index without recalculating existing scores. The methodology does not require retrospective revision when the city universe grows.

Anti-pattern resistant

The index is designed so that cities cannot easily game their scores by optimising for visible proxy metrics. The DI_PPP formula, AMPI penalty, and imbalance penalties are all difficult to manipulate without genuinely improving urban conditions.

11.

Notation Glossary

Symbol / Term	Definition
c	City or functional urban area being evaluated
m	Metric index (individual indicator within a pillar)
p	Pillar index (one of the five thematic pillars)
x_m(c)	Raw observed metric value for city c on metric m
s_m(c)	Normalized metric score for city c on metric m, range [0, 100]
μ	Arithmetic mean of normalized scores within a pillar or across pillars
σ	Population standard deviation of scores within a pillar or across pillars
cv	Coefficient of variation: σ / μ
AMPI	Adjusted Mazziotta–Pareto Index: $\mu - \sigma^2/\mu$
t_k	Fixed absolute anchor threshold at score k ($k \in \{0, 25, 50, 75, 100\}$)
DI_PPP(c)	PPP-adjusted monthly disposable income after essential costs for city c
α_(p,m)	Metric weight of indicator m inside pillar p
γ_m	Coverage weight for metric m (used in coverage ratio computation)
w_p	Public pillar weight (Growth 0.25, Viability 0.22, Capability 0.18, Community 0.15, Creative 0.20)
HAQ	Healthcare Access and Quality Index (IHME/Lancet)
GPI	Global Peace Index (Institute for Economics and Peace)
GII	Gender Inequality Index (UNDP)
GRI	Government Restrictions Index (Pew Research Center)
EGDI	E-Government Development Index (United Nations)
PPP	Purchasing Power Parity — conversion factor equalising purchasing power across currencies

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